Reader's Guide to the Relational Carrier Theory Corpus

(P0 — canonical synthesis of Papers 1–6 and the bridge note)

Sandro Bucciarelli*

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Abstract

This document is the canonical entry point to the relational-carrier-theory corpus. It is a synthesis, not a paper: every claim cites the load-bearing source paper, no new numerical result or proof is introduced here. The corpus consists of six sector papers (P1–P6) plus a bridge note; this guide states the theory's one-sentence thesis, identifies the three load-bearing claims, maps each paper to its role, registers the closure grammar (EXACT / PRECISE / FACTOR2; load-bearing / downstream / supporting / external / open), tabulates the top-tier closures, and lists the falsification handles that would settle the framework's main predictions within the next ~ 2 years. The intended reader is either a physicist arriving at the corpus for the first time or a reader wanting the architecture before the technical layers.

Preface: Why relations?

This corpus began from a simple question: what if physical reality is not built from objects placed inside space, but from relations that become stable enough to define space?

A difference by itself is not yet information. It becomes information only when it can be compared, reached, and conserved within a common structure. In this sense, relation is logically prior to information, and geometry to relation.

The technical work collected here explores this possibility in a reproducible way. The starting point is a finite relational carrier: a network of weighted relations whose structure is read through a bounded-operator basis and reduced to a small algebraic coefficient system. The claim tested throughout the corpus is not that the philosophical intuition is evidence, but that the same relational carrier can reproduce, constrain, or organise several physical sectors: emergent geometry, field-theoretic scales, loop-class corrections, black-hole entropy, dark-sector phenomenology, and falsifiable cosmological handles.

The philosophical role of this preface is therefore limited. It is not part of the proof structure. It does not introduce new assumptions, equations, or numerical claims. The technical claims begin in the Reader's Guide that follows and are owned by the individual papers. Each result stands or falls by its definitions, audits, reproducibility scripts, and falsification criteria.

The intuition behind the construction can be stated simply:

> *A world may be constituted not from things, but from reachable differences.*

If a difference cannot be reached, compared, or conserved, it is not yet physical information. If enough differences become globally reachable while remaining locally structured, a relational carrier can emerge. From such a carrier, geometry is not assumed as a background stage; it is something to be reconstructed. The Reader's Guide that follows maps the technical content of the corpus to specific load-bearing claims and falsification handles.

*Independent researcher. Correspondence: sandro.bucciarelli89@gmail.com.

What this document is and is not

What this is. A reader's guide. A map of where each question is settled in the corpus. A registry of the closure grammar. A pre-organised table of the load-bearing claims and falsification handles.

What this is not. Not a paper. Not a derivation. Not a declaration of completeness. Not peer-reviewed. The relational carrier theory is a work-in-progress closure construction by an independent researcher; the empirical and conceptual claims of Papers 1–6 are presented as candidate closures whose external validation is the principal next step. *This guide introduces no results of its own*: it is a navigational digest, so every numerical value, every closure form, and every falsification handle is reproduced from — and attributed to — the technical paper that derives and owns it. The original content lives in Papers 1–6; if a sentence in this guide and a sentence in the cited paper disagree, the cited paper is authoritative.

1 One-sentence thesis and short genealogy

One-sentence thesis. Physical reality is described as a finite relational carrier whose correlation structure generates the four observable channels of the bounded-operator readout — two primary (Ξ-cohesion, Ω-synchronisation) and two coupled (Ξ-ψ common, sync-floor) — from which the System-$\mathcal{R}$ rational coefficient tuple

$$(\gamma, \alpha_\xi, \varepsilon^2_{\text{sync}}, \beta_\pi, D_\Omega) \;=\; (1/10,\, 9/10,\, 1/20,\, 15/16,\, 67/80)$$

is read off, and from which the emergent geometry of general relativity, the Standard-Model fermion-mass-and-mixing spectrum, dark-sector phenomenology, and the Bekenstein–Hawking entropy $S/A = 1/4$ follow as algebraic consequences without additional free parameters.

Guiding principle. The whole construction can be read as a single extremal principle: *physical reality is the extremal organisation of realizable distinguishability* — the configuration that extremalises a structural action under finite reconstruction resources. This is not a separate postulate but a name for the variational backbone the corpus already carries: the seven-sector relational action of P6 [9] admits a variational principle whose stationarity conditions $\delta\mathcal{S}/\delta(\cdot) = 0$ slave every field to a deterministic lattice fixed point, and the emergent Einstein closure of P4 [4] is per-direction Galerkin residual minimisation. The guiding principle states what these constructions share; its content is exactly those published variational structures, not an additional axiom.

Foundational note: relation precedes information. The corpus's architecture instantiates a logical priority that is worth stating explicitly. A difference $a \neq b$ is not information in isolation: it requires a comparison relation in which a and b are mutually accessible. The relational carrier supplies that comparison structure as its primitive datum; the System-$\mathcal{R}$ rational tuple, the bounded-operator channels, and all downstream observables are read off from it *after* the relational structure is in place, not before. In this construction the name "relational carrier theory" is literal: *the relations come first, the differences become information by virtue of the relations*, and emergent geometry and physical observables are algebraic consequences of the relational correlation structure. The empirical Lemma B finding that the spectral gap λ_2 is globally distributed — not localised on any defect sub-population (P4 [4]) — is the operational counterpart of this priority: a connected relational carrier is the precondition for any observable to be *information about* anything else.

One-paragraph genealogy. The System-$\mathcal{R}$ primitive set is itself fully determined by two integer inputs — the relational-spacetime dimension $d = 4$ and the fermion generation count $N_{\rm gen} = 3$ — via $\gamma = 1/(2(d{+}1))$, $\alpha_\xi = (2d{+}1)/(2(d{+}1))$, $\varepsilon^2_{\rm sync} = 1/(4(d{+}1))$, $\beta_\pi = 15/16$, $D_\Omega = 1 - (N_{\rm gen}d{+}1)/(d^2(2N_{\rm gen}{-}1))$ (at the anchor $d{=}4, N_{\rm gen}{=}3$ the subtracted correction term is $(N_{\rm gen}d{+}1)/(d^2(2N_{\rm gen}{-}1)) = (N_{\rm gen}d{+}1)/(d^2(d{+}1)) = 13/80$, the two denominators agreeing via the algebraic identity $2N_{\rm gen}{-}1 = d{+}1$, so $D_\Omega = 1 - 13/80 = 67/80$; the Cabibbo-class form $(2N_{\rm gen}{-}1)$ is canonical, derived in Paper 4). Both d and $N_{\rm gen}$ are observationally fixed integer counts of discrete-geometric features of the underlying relational lattice, not free continuous parameters. The count of free continuous parameters in the structural input is zero; this is what the corpus means by "no free parameters". We emphasise the qualifier: $d{=}4$ and $N_{\rm gen} = 3$ *are* empirical inputs in the sense that observation of our world fixes their values, and the framework does not derive these integers from a deeper structure. This corpus is a *minimal-input program*: two integer inputs and one additional structural anchor (the carrier-action functional form $S_{\rm UV}$), all other parameters derived. It is not a "zero information input" programme in the sense of a theory that derives spacetime dimensionality and generation count from first principles — that remains a separate analytical target. Readers comparing this corpus to "fully derived" fundamental theories should bear this distinction in mind: RCT is parameter-free in the continuous sense, not input-free in the discrete-geometric sense. The five System-$\mathcal{R}$ rationals are the algebraic backbone of all downstream closures.

Derived primitive identities. Several closed-form identities among the primitives recur across the corpus and are used repeatedly in the downstream closures. For convenience of cross-paper reading they are registered here:

- $\alpha_\xi + \gamma = 1$ (chirality complementarity; P2 § 5);
- $\varepsilon^2_{\rm sync} = \gamma/2$ (fluctuation–dissipation symmetry; P2 § 5);
- $D_\Omega = \beta_\pi - \gamma = 15/16 - 1/10 = 67/80$ (diffusion–projection balance; the closed-form definition of D_Ω among the primitives, closed by theorem in P6 [9] and the C2 constraint of P2 § 5);
- $2N_{\rm gen} - 1 = d + 1 = 5$ (Cabibbo-class identity; makes the two D_Ω genealogy denominators agree and sets the V_{cb} dimensional structure);
- $\gamma^2 = 1/100 = \big(2(d{+}1)\big)^{-2}$ (the universal second-order "leakage" scale: it sets the γ^2 prefactor of $n_s{=}1 - \gamma^2(d{+}N_{\rm gen})/2$, the $\gamma^2 c_X$ sub-leading corrections of P4 [4], and the α_ξ^2 vs. $\alpha_\xi^2(1{-}\gamma^2)$ Continuum-Backreaction-Identity ambiguity);
- $\alpha_\xi^2 = 81/100$ (the variance-reactivity / backreaction continuum value: it recurs as the common leg-value of the Continuum Backreaction Identity and as the within-P_5 T^{Ξ}_{00} Symanzik asymptote in P4 [4]);
- $s_{\rm face} = 1/d = 1/4$ (Bekenstein–Hawking entropy face, equivalently the inverse-dimension prefactor; P3 L7);
- $d + N_{\rm gen} = 7$ (mass-mode dimensional primitive; recurs in $L4 = -(d{+}N_{\rm gen})^2/(d(d{+}1)) = -49/20$, $n_s{=}1 - \gamma^2(d{+}N_{\rm gen})/2{=}193/200$, and $Y_p{=}(d{+}N_{\rm gen})^2\gamma^2/2{=}49/200$);
- $2(d{+}1) = 10 = \gamma^{-1}$ (back-channel inverse; the leading integer in the Coleman–Gamow action $S_{\rm Gamow}{=}2(d{+}1)N_{\rm gen} - (2d{+}N_{\rm gen}){=}19$);
- $2d + N_{\rm gen} = 11$ (CKM dimensional primitive; $V_{cb}{=}\alpha_\xi/(2(2d{+}N_{\rm gen})){=}9/220$);

- $(d+1)/d = 5/4$ (gravitational-backreaction screening amplitude; cosmological-constant Schwinger–Dyson layer);
- $N_{\text{gen}} \cdot d = 12$ (occupancy-sector dimensional product; pairs-per-cell on the canonical ladder).

Each identity is a closed-form consequence of the integer inputs (d, N_{gen}); they are not independent free parameters.

2 Three load-bearing claims

The technical-foundations paper P6 [9] states the load-bearing classification explicitly in its "three claims are load-bearing" paragraph: the construction stands on

1. the *bounded carrier with finite-N Einstein-residual closure* of P4 [4] (emergent gravity, lattice $\Lambda_{\text{lat}} = 19/15$ identity, four-point Continuum Backreaction Identity);
2. the *System-$\mathcal{R}$ coefficient readout via the bounded-operator construction* of P2 [2] (ten causal-wave landings, formula-search audit, the four-projector spectral uniqueness theorem);
3. the *topology-labeled loop-class library mapping* of P3 [3] (loop-class protocol, Yukawa-cluster PROSP-01/04, Bekenstein–Hawking $S/A = 1/4$).

All other cross-sector closures — α_{EM}, v_{EW}, $S_{\text{BH}}/A = 1/4$, n_s, A_s, halo phenomenology, dark sector — are *downstream* of the three load-bearing claims in the sense defined in Section 5.

3 Two branches: vacuum and matter-side chirality flip

The bridge note [10] formulates the corpus-wide two-branch structure as a single *loop-topological carrier* operating in two distinct relational regimes:

Vacuum branch. Pre-chirality-flip configuration ($\theta_{\text{chir}} = 0$, canonical $(\alpha_\xi, \gamma) = (9/10, 1/10)$). Governs the electroweak-scale closure ($v_{\text{EW}} = 246.2186$ GeV via P1 [1]'s four-path consensus), the inflation-era and recombination-era observables $(n_s, A_s, z_*, \theta_*)$, and the late-time dark-energy equation of state at leading order ($w_{\text{DE}} = -39/40$).

Matter-side chirality-flip branch. Post-$\pi/4$ matter-side branch with asymptotic/inverted anchor $(\alpha_\xi, \gamma) = (1/10, 9/10)$: the chirality-flip boundary itself sits at $\theta_{\text{chir}} = \pi/4$ (where the two coefficients cross), and the asymptotic post-flip anchor is the inverted tuple $(1/10, 9/10)$ approached as θ_{chir} continues past $\pi/4$. This branch governs the matter-core phenomenology (top-5% K/Q closure), the chirality-running $w_{\text{DE}}(\theta_{\text{chir}})$ at next-to-leading order (P4B [6]), and the absolute neutrino-mass identification $m_1 = \gamma^2 m_3$ via the loop-class companion's Pure-Self-Energy resummation (P3 [3]).

The two-branch decomposition is itself System-$\mathcal{R}$-derived rather than ad hoc: the chirality angle $\theta_{\text{chir}}(N)$ runs algebraically with lattice resolution N [4] (the running formula is the one labelled *theta-running-first-principles* in P4), crossing $\pi/4$ at a specific lattice resolution at which the back-channel-projection numerical hierarchy inverts.

Paper	Title (short)	Role in the corpus
P0 (this doc.)	Reader's guide	Entry-point, claim grammar, corpus map, falsifiers
P1 [1]	Hierarchy-bridge electroweak-scale closure	$v_{\rm EW} = 246.2186$ GeV four-path consensus, $\lvert g\rvert_{\overline{\rm MS}} = 1.4187$, HBR + $\theta_{em} = \pi$ load-bearing input to P4
P2 [2]	Causal-wave landings + System-$\mathcal{R}$ readout	**Pillar (ii)**: bounded-operator construction; ten landings; four-projector spectral-uniqueness theorem; formula-search audit 47/$\sim$174k
P3 [3]	Loop-class library + closure protocol	**Pillar (iii)**: loop-class topology-labelled mapping; Bekenstein–Hawking $S/A = 1/4$; SYE Yukawa/CKM/PMNS pipeline; PROSP-01/04 prospective registry
P4 [4]	Emergent-gravity bulk Einstein closure	**Pillar (i)**: bounded carrier; $\Lambda^{\infty}_{\rm lat} = 19/15$; Continuum Backreaction Identity; closure-domain ladder $N \in [50, 512]$
P4A [5]	Schwarzschild + PPN companion	Companion test: weak-field PPN parameters; $\bar{\theta}_{\rm QCD}$ vanishing
P4B [6]	Anisotropic source $\to$ DM/DE companion	Companion: $T^{\Xi}_{\mu\nu}$ decomposition; $w_{\rm DE} = -39/40$; chirality-running $w(z)$; 9-layer cosmological-constant dressing
P4C [7]	H3c witnesses companion	Companion diagnostics for the Einstein-residual closure
P4D [8]	Atiyah–Singer chirality index companion	Companion: integer-quantised vortex-DM winding number; topological-DM identification
Bridge note (appendix) [10]	P1$\leftrightarrow$P4 cross-file consistency	Lemma-level: common loop-topological carrier; version-lock matrix; cross-paper $\mathcal{R}$-tuple consistency
P6 [9]	Relational-UV-closure technical foundations	Master technical document: H_{UV}, S_{UV}, Z_{UV}, branch construction, closure ledger, load-bearing-vs-downstream registry

Table 1: Corpus map. P0 (this document) is the reader's guide and entry point; P2, P3, P4 are the three load-bearing claims of Section 2; P1, P4A–D, and P6 supply supporting structure or technical foundations; the bridge note is a cross-file consistency appendix.

4 Corpus map: which paper does what

5 Claim grammar

The corpus uses a registered vocabulary of closure tiers and claim classes. Each is defined formally in one of the technical papers; the present section is a routing table only.

Closure tiers (precision against external anchor).

- **EXACT**: relative residual at or below the paper-specific EXACT cut, corresponding to a System-$\mathcal{R}$ identity that closes at machine precision under the rational reduction $\mathcal{R}$ (see P6 §1 definition).
- **PRECISE**: relative residual $\leq 2.5\%$; closes the framework prediction within standard-fit experimental uncertainty.
- **FACTOR2**: closure form correct but numerical value within a factor of 2; reported as a structural-form identification, not a precision claim.
- **STRUCTURAL_CLOSED**: closure form derived algebraically; numerical match secondary (e.g. $\Sigma_f Y_f^2 = 10$).

Terminological disclaimer on the word "EXACT". Throughout this corpus the labels EXACT, PRECISE and FACTOR2 are *band labels*, not claims of mathematical equality. EXACT here means "the framework value lies inside the indicated relative-residual band against the external anchor"; a 1% residual against an experimental measurement is not exact in the mathematical sense of $a = b$ in $\mathbb{Q}$. Where the corpus does report a true algebraic identity in $\mathbb{Q}$ (e.g. $\alpha_\xi/2 - 2\gamma = 1/4$ for S_{BH}/A, or $\Sigma_f Y_f^2 = 10$ for the hypercharge-squared sum rule), the closure is flagged as **STRUCTURAL_CLOSED** or written with the explicit identity symbol $\equiv$ in the body text. Readers comparing the corpus against algebraic-identity theories (e.g. representation-theoretic constructions in $\mathbb{Q}$) should treat EXACT-band labels as "Tier-1 closure within band", not as algebraic identities.

We use Tier-1 / Tier-2 / Tier-3 interchangeably with EXACT / PRECISE / FACTOR2; the Tier-k vocabulary is preferred when the rhetoric of EXACT could mislead an external reader who expects mathematical equality.

Paper-specific EXACT cuts. The corpus uses three EXACT cuts depending on the observable class. The present routing table makes them explicit so a reader cross-comparing two papers does not confuse a stricter EXACT-label in one paper with a looser one in another. A claim that is EXACT under a stricter cut is automatically EXACT under any looser cut (the implication runs from left to right).

Paper / observable class	EXACT cut	Rationale
P2 (causal-wave landings)	$\lvert r\rvert \leq 0.01\%$	EXACT-target algebraic landings; sub-experimental-uncertainty closure
P3 (loop-class library)	$\lvert r\rvert < 0.4\%$	loop-corrected observables compared against PDG/NuFIT measurement bands
P0 / P4 / P4A–D / P6 / bridge	$\lvert r\rvert \leq 1\%$	global manifest band covering all observable classes; superset of P2 and P3 cuts

The PRECISE cut is uniform across the corpus ($|r| \leq 2.5\%$) and the FACTOR2 cut is uniform ($|r| \leq 100\%$, i.e. within a factor of two). A claim that closes at $|r| \leq 0.01\%$ is EXACT under any of the three cuts; a claim at $|r| \in (0.01\%, 0.4\%]$ is EXACT under P3, P0 and the global manifest but downgrades to PRECISE under P2; a claim at $|r| \in (0.4\%, 1\%]$ is EXACT under P0 and the global manifest but PRECISE under P3.

Claim classes (load-bearing classification). The unified manifest (file `corpus_claims_manifest.json` in P6's repository [9]) registers every main claim under one of five labels:

- **Load-bearing.** The construction stands on this claim. Three exist: pillars (i)–(iii) of Section 2.
- **Downstream.** An algebraic or audit consequence of the same System-$\mathcal{R}$ tuple read off by the load-bearing claims (e.g. $\alpha_{\mathrm{EM}} = \gamma^2 \alpha_\xi^{N_{\mathrm{gen}}}$, $\sigma_8 = \alpha_\xi^2$, $n_s = 1 - 7\gamma^2/2$). The vast majority of the closures of Table 2 fall here.
- **Supporting witness.** An independent audit diagnostic that does not extend the closure surface but confirms it (e.g. P4C's H3c gravitational-coupling-scaling diagnostics).
- **External contact.** A point of contact with the existing literature (Pauli–Zeldovich cancellation, Schwinger–Dyson resummation, Reed–Simon VIII.5).
- **Open target.** A registered observable whose closure is not yet established within the corpus and which is flagged as the framework's primary near-term theoretical target.

6 Main closures at a glance

The list below is a curated subset of the ~ 50 registered closures of the corpus, selected for highest external-anchor visibility. Each entry is owned by the cited paper; this table re-states them in one place.

Table 2: Top-tier corpus closures, one row per observable. "Tier" is the precision class against the external anchor: **EXACT** ($\equiv$ Tier-1; $|r| \leq 1\%$), **PRECISE** ($\equiv$ Tier-2; $|r| \leq 2.5\%$), **FACTOR2** ($\equiv$ Tier-3; within $2\times$). All three are band labels, not claims of mathematical equality; see the terminological disclaimer in Sec. 5. "Class" is the load-bearing classification of Section 5. Rows marked *load-bearing* are components of the three load-bearing blocks (pillars (i)–(iii) of Section 2), not additional independent pillars; the corpus has *exactly three* load-bearing pillars and the multiplicity of load-bearing rows in the table reflects the multiple observables that each pillar supplies.

Observable	System-$\mathcal{R}$ form	Tier	Class	Owner
		Cosmology		
H_0	$100\,\alpha_\xi\, s_{\mathrm{face}}\, N_{\mathrm{gen}} = 67.5$ km/s/Mpc	EXACT	downstream	P4B
σ_8	$\alpha_\xi^2 = 81/100$	PRECISE	downstream	P4B
S_8	$\alpha_\xi^2\sqrt{\Omega_m/0.3} = 0.828$	EXACT	downstream	P4B
Ω_m	$47/150$	PRECISE	downstream	P4B
Ω_Λ	$103/150$	PRECISE	downstream	P4B
$\Omega_{\mathrm{DM}}h^2$	$243/2000 = 0.1215$	PRECISE	downstream	P4B
w_{DE}	$-1 + \varepsilon_{\mathrm{sync}}^4/\gamma = -39/40$	PRECISE	downstream	P4B

Observable	System-$\mathcal{R}$ form	Tier	Class	Owner
n_s	$1-7\gamma^2/2$	EXACT	downstream	P4B
A_s	$21\gamma^{10}$	EXACT	downstream	P4B
z_*	$(2d{+}1)(2d{+}N_{\rm gen})^2{+}$sub	EXACT	downstream	P4B
Y_p	$49\gamma^2/2$	EXACT	downstream	P4B
Σm_ν (NH-min, dressed)	$m_2^{\rm dr}+m_3^{\rm dr}=0.0588$ eV; seesaw refinement $\gamma^2 m_3{+}m_2{+}m_3\simeq 0.0593$ eV	EXACT	downstream	P4B
$m_3^{(\nu)}$ atmospheric	$m_e\gamma^{d+3}(1-\gamma^2(d{+}N_{\rm gen})/4)=0.0502$ eV	EXACT	downstream	P4B
$m_2^{(\nu)}$ solar	$m_e\gamma^{d+4}(\pi/2)(1+\gamma^2(d{+}N_{\rm gen}))=0.00859$ eV	EXACT	downstream	P4B
η_B	$(d{+}N_{\rm gen})^2\gamma^{10}/(2d)=49/(8{\cdot}10^{10})$	PRECISE	downstream	P4B
D/H	$(d{+}1)\gamma^5/2\,(1+\gamma^2(d{+}N_{\rm gen})/4)=2.544\times10^{-5}$	EXACT	downstream	P3
	Electroweak & Standard Model			
$v_{\rm EW}$	4-path consensus 246.2186 GeV	PRECISE	load-bearing input	P1
$\alpha_{\rm EM}$	$\gamma^2\alpha_\xi^{N_{\rm gen}}=729/100{,}000$	EXACT	downstream	P3
$\sin^2\theta_W$	$1/d-\gamma/(2N_{\rm gen})=7/30$	PRECISE	downstream	P3
V_{us}	$\alpha_\xi\, s_{\rm face}=9/40$	EXACT	downstream	P3
m_t	spectral Yukawa eigenvalue $\to$ 174.1 GeV	EXACT	downstream	P3
m_μ/m_e	$400N_{\rm gen}(d{+}1)/P_{29}=6000/29=206.897$	EXACT	downstream	P3
$a_\mu^{\rm Schwinger}$	$\alpha_{\rm EM}/(2\pi)$	PRECISE	downstream	P3
	Gravity & black-hole sector			
$\Lambda_{\rm lat}^\infty$	$3\alpha_\xi/2+\varepsilon_{\rm sync}^2-\gamma(N_{\rm gen}{+}1)/N_{\rm gen}=19/15$	EXACT	load-bearing	P4
Λ_t^∞	$\alpha_\xi^2=81/100$	EXACT	load-bearing	P4
Λ_s^∞	$-\gamma^2/2=-1/200$	EXACT	load-bearing	P4
$S_{\rm BH}/A$	$s_{\rm face}=1/d=1/4$	EXACT	load-bearing	P3
PPN $\gamma_{\rm PPN}$	induced from Einstein closure	PRECISE	downstream	P4A
$\rho_\Lambda^{\rm pred}/\rho_\Lambda^{\rm obs}$	9-layer dressing $\to$ 1.001	EXACT	downstream	P4B
	Dark sector & halo phenomenology			
$g_\dagger$ (RAR)	$cH_0/(2\pi\alpha_\xi)$	PRECISE	downstream	P4B

Observable	System-$\mathcal{R}$ form	Tier	Class	Owner
BTFR A	$\sim(G_N H_0)^{-1}$	PRECISE	downstream	P4B
NFW vs Burkert branching	polarity-violation rate	STRUCTURAL	downstream	P4B
SPARC pass-fraction	79/129 within domain $\mathcal{D}_{\rm halo}$	PRECISE	downstream	P4B
Vortex-DM $w_{\rm DM}$	0 (integer-quantised winding)	STRUCTURAL	downstream	P4D
		Foundations		
4-projector basis	Reed–Simon VIII.5 spectral-uniqueness	EXACT	load-bearing	P2
10 causal-wave landings	47 forms / $\sim$ 174k searched	EXACT	load-bearing	P2
Loop-class library	topology-labelled protocol	STRUCTURAL	load-bearing	P3
$\sum_f Y_f^2 = 10$	three-generation chiral content	EXACT	downstream	P6
$\bar{\theta}_{\rm QCD} = 0$	charge-neutral vortex/antivortex condensate (P4 + P4D + P3 chirality-pair)	STRUCTURAL	downstream	P4D
Emergent causal time / 3+1 Lorentz layer	causal partial order on Ξ-paths + massless transport light cone; $d_{\rm spatial} = 3$, $d_{\rm spacetime} = 4$, $f_{\rm LC} = \log_2(9/8) = 0.169925$	STRUCTURAL	supporting	P4
Emergent UV time t (Page–Wootters)	t = Fiedler-mode-phase clock eigenvalue of Ξ-Laplacian; $\hat{C}\lvert\Psi\rangle = 0$ Wheeler–DeWitt constraint	STRUCTURAL	supporting	P6

7 Falsification handles

The framework registers explicit falsification triggers against current and near-term cosmological and laboratory data. Closure of these handles within the next $\sim$2 years is the principal external test of the framework.

Cross-reference to owning-paper falsifier IDs. The six corpus-wide falsifiers below map to the owning-paper local falsifier labels as registered in each companion paper's `Falsification triggers` section.

P0 falsifier (this section)	Owning paper	Local label
1. Σm_ν vs DESI / CMB-S4 / Roman / Euclid	P3, P6	P3 FAL-Σm_ν, P6 §FAL-cosmo
2. Dynamical $w(z)$ vs DESI BAO running	P4B	P4B FAL-w_a
3. Hubble tension on non-Cepheid distance ladders	P4B	P4B FAL-H_0
4. Tensor-to-scalar ratio r vs CMB-S4	P4B	P4B FAL-r
5. Particle-DM direct detection (LZ / XENON-nT / PandaX / DarkSide)	P4D	P4D F1, F2, F3
6. Halo regime signatures (microlensing, EDGES 21-cm)	P4, P4B	P4B FAL-halo

A reader following one of these falsifiers across to its owning paper finds the same numerical thresholds at higher technical detail and the reproducer pointers.

Cosmological-data-decidable within $\sim$2 years.

1. *Σm_ν vs DESI / CMB-S4 / Roman / Euclid.* The framework predicts $\Sigma m_\nu \simeq 0.060$ eV via $m_1 = \gamma^2 m_3$ [3, 9]. The current DESI DR2 BAO + DR1 full-shape bound is $\Sigma m_\nu <$ 0.0642 eV [11]; the framework prediction sits 6.1% below this. A confirmed tightening below the NH oscillation minimum $\simeq$0.0586 eV would falsify the $\gamma^2 m_3$ identification.

2. *Dynamical $w(z)$ vs DESI BAO running.* The framework predicts $|w_a| \lesssim \gamma^2 = 0.01$ from the chirality-running structure $w_{\rm DE}(\theta_{\rm chir})$ [6]. The DESI 2024+2025 central $w_a \approx -0.4$ at $\sim 3\sigma$ lies $\sim 40\times$ outside the framework band; consolidation to $>5\sigma$ in DESI-DR3 would falsify the $\varepsilon^4_{\rm sync}/\gamma$ closure.

3. *Hubble tension on non-Cepheid distance ladders.* The framework predicts $H_0 =$ 67.5 km/s/Mpc independent of distance-ladder method [6]. Confirmed $H_0 \in [72, 74]$ km/s/Mpc on a non-Cepheid local ladder (gravitational-wave sirens, TDCOSMO, JWST-anchored TRGB at $\sigma_{H_0} \lesssim 1$ km/s/Mpc) would falsify the distance-ladder-systematic attribution of the SH0ES 73.04$\pm$1.04 discrepancy.

4. *Tensor-to-scalar ratio r vs CMB-S4.* The framework predicts $r \sim \gamma^3 (d + N_{\rm gen}) \simeq 7 \times 10^{-3}$ within the end-of-2025 Planck–SPT–ACT–BICEP/Keck–DESI 95% bound $r < 0.034$ [13] (subsuming the earlier BK18 bound $r_{0.05} < 0.036$ [12]). CMB-S4 sensitivity $\sigma(r) \sim 10^{-3}$ within $\sim$3 years will provide a definitive detection or upper-bound test.

Laboratory-data-decidable.

5. *Particle-DM direct detection.* The framework identifies dark matter with a relational vortex-defect network of integer-quantised $U(1)$ winding [8]; this is topological, not particulate. The framework therefore predicts *null* results on LZ ($\sigma_{\rm SI} < 2.2 \times 10^{-48}\,{\rm cm}^2$ at 40 GeV/c^2 [14]), XENON-nT [15], PandaX-4T [16], and DarkSide [17] down to the neutrino floor. A positive nuclear-recoil signal at any future facility would falsify the topological-DM identification.

6. *Halo regime signatures.* Microlensing of vortex-network clumps at the chirality-flip mass scale $M_{\rm flip}$ [4] should show a discrete event-frequency distribution rather than a continuous PBH spectrum; 21-cm absorption-depth deviations from ΛCDM in the EDGES band correlate with the vortex-density power spectrum at $z \sim 17$.

Pre-registered prospective targets. The loop-class companion paper P3 [3] registers a prospective cluster registry — the file `prospective_cluster_registry.json` in the loop-class repo — that pre-registers three loop-class observables for future $\mathcal{R}$-falsification opportunities at the P3 protocol level.

Composition theorem conditionality: the (SG) **axiom.** The composition theorem of P6 [9] states that $(M1)$, $(M2)$, $(M4)$ are unconditional theorems of the cited papers, while $(M3)$ — the emergent-Einstein closure — is *conditionally rigorous* on three inputs: (i) the eight admissibility conditions A_1–A_8 on the carrier sequence, (ii) the discharge-table implication $(A_5)+(A_6)\Rightarrow CD(K_{\mathrm{CD}}, N)$ open at theorem level in P4, and (iii) the empirical Symanzik asymptotes underlying the Continuum Backreaction Identity (T^{Ξ}_{00} leg $R^2 = 0.54$, auxiliary $G_{00,\mathrm{med}} \to 0$ at $R^2 = 0.99$). The P6 sharpening reduces (i) to a *single* structural axiom: the uniform spectral gap (SG) on the ξ_N-weighted normalised graph Laplacian, with branched asymptotes $\lambda_\infty^{(\mathrm{vac})} = 3/8 = (d-1)/(2d)$ (vacuum, Bayesian posterior 0.908 on the $N \le 100$ ladder) and $\lambda_\infty^{(\mathrm{mat})} = 79/200 = 3/8 + d\gamma^2/2$ (matter-stable, $256 \le N \le 600$, -0.16% relative residual, Bayes-factor $S_2/S_1 = 0.999$). Both closures hold unconditionally on the canonical carrier sequence: the chirality-running family $\theta_{\mathrm{chir}}(N) = \arctan(N_{\mathrm{gen}}^{2x-1})$ with $x = \ln(N/N_{\mathrm{canonical}})/\ln(dN_{\mathrm{gen}})$ (P4B [6]). The fixed-input-parameter test ladder used for empirical certification realises the canonical sequence up to the chirality-inversion endpoint $N_{\mathrm{inv}} = dN_{\mathrm{gen}}N_{\mathrm{canonical}} = 600$; extension of the empirical certification to higher N requires the standard lattice-continuum parameter rescaling that preserves θ_{chir} on its canonical trajectory. The fixed-parameter $N > N_{\mathrm{inv}}$ regime exits the canonical sequence (chirality saturates at $\arctan(N_{\mathrm{gen}})$, lattice exhibits uniform ξ-densification with a drifting non-asymptotic spectral gap; P4 [4] Remark `rmk:longer_iter_accelerating`) and is a numerical artefact of fixed-parameter ladder construction, not a falsification of the (SG) axiom. The astrophysical and cosmological observable scales of the corpus realise the canonical sequence and are unaffected. Neither branch is analytically discharged at theorem level. A GPU decomposition study (P4 Prop. Statistical-fingerprint, 2026) further reduces the open analytical question to a concrete graph-theoretic statement: the spectral gap of the Newman-clustered ensemble with the carrier's empirical (degree, triangle) distribution equals 7/24 at $+0.36\%$ relative residual — and *only* this ensemble reproduces the carrier asymptote (Erdős–Rényi, configuration model, random-regular, and Watts–Strogatz all miss by -40 to $+66\%$). The corresponding analytical tool is the cavity / resolvent method for sparse networks with triangles; the remaining Step-4 deliverable is therefore concretely the closed-form cavity fixed-point solution for the carrier's clustered-degree distribution. *Falsification handle*: any rigorous proof that the carrier-clustered-graph cavity solution does *not* yield 7/24, or any tightened empirical measurement that excludes 3/8 on a $\mathcal{R}$-aware ladder at $\ge 5\sigma$, would close the axiom against the framework rather than for it.

Physical mechanism of the corpus closures. Each closure of the corpus is classified along two independent axes: a *numerical-precision tier* (EXACT / PRECISE / FACTOR2; the closure-grammar registry above) and a *physical-mechanism status* (rigorous derivation, physical plausibility argument, or open). Every numerical landing in the corpus comes with either a rigorous physical derivation or a concrete plausibility argument connecting the System-$\mathcal{R}$ form to the observable; no closure is registered as purely numerical without physical reading.

A subset of the closures admit no single-route derivation but are nonetheless established by a *cross-projection argument*: two or more independent in-corpus chains connect to the observable through standard physics, and their numerical predictions converge at the PRECISE tier. The three cosmological and geometric closures listed below are of this kind.

- *Local Hubble constant $H_0 = 67.5$ km/s/Mpc.* The dimensionless reading $h := H_0/(100\,\mathrm{km/s/Mpc}) = \alpha_\xi\, s_{\mathrm{face}}\, N_{\mathrm{gen}} = 27/40$ from the carrier-side product is also delivered by the standard Friedmann equation at $z = 0$, with the framework's independent dressing of ρ_Λ [6] and its closure $\Omega_\Lambda = 1 - \Omega_m = 103/150$ as inputs. The carrier-side route gives 67.50 and the Friedmann route 66.93; the two converge at 0.85% relative residual, each within Planck 2018 1σ of the empirical 67.4 ± 0.5.
- *Lattice cosmological-constant asymptote $\Lambda^\infty_{\mathrm{lat}} = 19/15$.* The operator-side decomposition

$17/20+5/12$ (non-scalar Clifford-channel reaction rate plus spinor-trace–generation correction, [4]) is algebraically equivalent to the carrier-side decomposition $1+d/((d+1)N_{\text{gen}}) = 1+4/15$ (trivial unit carrier-baseline plus per-generation chirality-spin defect distributed over the sync-extended cone bundle). Two independent physical projections of the same limit, agreeing identically at 19/15.

- *Curvature-dimension lower bound* $\mathrm{CD}(K_{\text{CD}} \geq 0, N)$. Three independent Bakry–Émery percentile statistics each match a $\lambda_\infty \times (d, N_{\text{gen}})$-rational — $K_{p_1} = \lambda_\infty(d+1)/d = 15/32$ at 0.19% residual, $K_{p_5} = \lambda_\infty N_{\text{gen}}(d+N_{\text{gen}})/2^d = 63/128$ at 0.04% residual, $K^{\text{mean}} = \lambda_\infty(4d+1)/(2d+N_{\text{gen}}) = 51/88$ at 0.10% residual. The three-fold convergence identifies the Bakry–Émery curvature distribution as a structural function of the (SG)-axiom λ_∞ and the integer inputs (d, N_{gen}), extending the qualitative Ollivier–Lin–Lu–Yau corollary $K_{\text{CD}} \geq 0$ to a quantitative cross-projection.

The cross-projection arguments are not stand-alone derivations: each identity rests on standard physics (Friedmann; Bakry–Émery Γ_2-calculus; algebraic-decomposition equivalence) plus the framework's prior closures of the routes' inputs. The load-bearing evidence is the empirical convergence of two or more independent in-corpus chains at residuals of 0.04–0.85%, all inside the bootstrap 95% confidence intervals of the corresponding empirical measurements. The single remaining mechanism conditionality of the corpus is therefore the discharge of the (SG) axiom $\lambda_\infty = 3/8$ itself: a closed-form derivation would, via the cross-projection chains above, simultaneously deliver rigorous physical mechanisms for all three closures of this kind.

Statistical multiplicity / look-elsewhere. The corpus reports many closures, which invites a look-elsewhere objection. The defence does *not* rest on post-hoc p-value-inflation corrections; it rests on five structural features that a random-search artefact would not exhibit. *(i) Dependency graph*: the closures are not independent fits but a directed graph rooted in the two integers (d, N_{gen}) — every System-$\mathcal{R}$ rational is a fixed algebraic function of those two inputs, so the effective number of free choices is two, not the number of closures. *(ii) Frozen readouts*: the five carrier coefficients are read off a bounded-operator construction (P2 [2]) once and then frozen; no downstream closure re-tunes them. *(iii) Negative controls*: the loop-class protocol (P3 [3]) has explicit forbidden classes and a closure domain outside which the protocol is *required* to fail, and several candidate identifications have been falsified on those controls rather than retained. *(iv) Prospective registries*: P3 pre-registers loop-class observables (`prospective_cluster_registry.json`) before their external values enter the closure, converting them from fits into predictions. *(v) No target-fitting*: the System-$\mathcal{R}$ forms carry no fitted parameter — the admissible alphabet and the formula-search audit of P2 [2] are disclosed, and a closure either lands on a System-$\mathcal{R}$ rational or it does not. A look-elsewhere artefact would require all five to hold by coincidence; the framework is therefore best contested through the registered falsification handles above, not through a multiplicity correction on a closure count.

8 Where to read next

The natural reading order depends on the reader's question:

"What is the theory?". Read this guide (P0) first. Then read P6 [9] for the technical foundations, P2 [2] for the System-$\mathcal{R}$ readout, P3 [3] for the loop-class protocol, and P4 [4] for the main emergent-GR claim. The bridge note [10] is best read afterward as a cross-file consistency appendix.

"Where do the System-$\mathcal{R}$ coefficients come from?". P2 [2] (causal-wave landings + formula-search audit + spectral-uniqueness theorem).

"How does general relativity emerge?". P4 [4] (bounded carrier + Einstein-residual closure + Continuum Backreaction Identity); then P4A [5] for the weak-field PPN companion.

"How do the Standard-Model masses and mixings close?". P3 [3] (loop-class library + SYE Yukawa-pipeline + α_{EM}-mediated identities).

"How do dark matter and dark energy fit in?". P4B [6] (anisotropic source decomposition, w_{DE}, cosmological-constant dressing) and P4D [8] (vortex-DM topological identification).

"What is the electroweak-scale closure?". P1 [1] (HBR + $\theta_{em} = \pi$ four-path consensus for v_{EW}).

"Is the corpus internally consistent?". The bridge note [10] (technical-consistency proof P1↔P4) and the unified manifest file `relational-uv-closure-repro/data/corpus_claims_manifest.json` of P6 [9].

Closing note

The relational carrier theory is presented as a candidate closure construction whose external validation is the principal next step. The corpus does not claim to be a final accepted theory of nature; it claims that the load-bearing pillars (i)–(iii) of Section 2 are internally consistent, that the downstream closures of Table 2 follow algebraically from the load-bearing pillars within the System-$\mathcal{R}$ rational reduction, and that the falsification handles of Section 7 provide a decidable near-term test surface. Each individual claim is owned by the cited paper; the present guide is an entry point and a map, nothing more.

Data and software availability

The reproducibility package (`https://github.com/TerraSignum/relational-carrier-manifest`) bundles all input data files (under `data/`), reproducer scripts (`src/verify_*.py` and `make_figures`), and unit tests (`tests/`). Cryptographic provenance is provided by `data/SHA256SUMS` (GNU `sha256sum -c` compatible), and the publication-prep frozen state is documented in `RELEASE_v1_0_0.md` with full release notes in `CHANGELOG.md`. Software metadata is in `pyproject.toml` and `CITATION.cff` (Citation File Format 1.2). The package is licensed under MIT; the manuscript text is licensed under CC-BY-4.0.

References

[1] Bucciarelli, S., *A defect-field two-loop correction to the Hierarchy-Bridge-Relation electroweak scale* (`hbr-ew-scale-repro`, 2026).

[2] Bucciarelli, S., *A five-coefficient causal-wave ansatz and ten cross-sector benchmark closures* (`causal-wave-landings-repro`, 2026).

[3] Bucciarelli, S., *Topology-labeled loop-class mappings for cross-sector physical observables* (`loop-class-closure-repro`, 2026).

[4] Bucciarelli, S., *Emergent Einstein dynamics from relational metric closure on finite correlation lattices* (`emergent-gr-closure-repro`, 2026).

[5] Bucciarelli, S., *Schwarzschild defect, post-Newtonian limits, and the black-hole sector from a relational emergent-metric construction* (`emergent-gr-schwarzschild-ppn-repro`, 2026).

[6] Bucciarelli, S., *Anisotropic source tensor and the dark-matter / dark-energy mixture decomposition on a relational lattice* (`emergent-gr-anisotropic-source-dm-de-repro`, 2026).

[7] Bucciarelli, S., *Multi-observable indirect-witness chain bounding the Einstein-identity gap on a relational emergent-gravity lattice* (`emergent-gr-h3c-witnesses-repro`, 2026).

[8] Bucciarelli, S., *A discrete Atiyah–Singer-type bridge from the chirality-balance witness to the Einstein-identity gap on the relational Ξ-graph* (`emergent-gr-atiyah-singer-chirality-repro`, 2026).

[9] Bucciarelli, S., *Full UV-closure construction of the relational QFT–Einstein carrier: a two-phase loop-topological framework with vacuum and matter branches* (`relational-uv-closure-repro`, 2026).

[10] Bucciarelli, S., *A common loop-topological carrier for electroweak and emergent-gravity sectors (Technical consistency note across Papers 1–4)* (`qft-einstein-bridge-repro`, 2026).

[11] W. Elbers *et al.* (DESI Collaboration), *Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape*, Physical Review D **112**, 083513 (2025); arXiv:2503.14744.

[12] P. A. R. Ade *et al.* (BICEP/Keck Collaboration), *Improved constraints on primordial gravitational waves using Planck, WMAP, and BICEP/Keck observations through the 2018 observing season* (BK18), Physical Review Letters **127**, 151301 (2021); arXiv:2110.00483.

[13] L. Balkenhol *et al.*, *Inflation at the End of 2025: Constraints on r and n_s Using the Latest CMB and BAO Data*, arXiv:2512.10613 (December 2025). Combined Planck–SPT–ACT–BICEP/Keck + DESI BAO analysis; $n_s = 0.9682 \pm 0.0032$ (CMB only), 0.9728 ± 0.0029 with BAO; 95% C.L. upper bound $r < 0.034$.

[14] J. Aalbers *et al.* (LUX-ZEPLIN Collaboration), *Dark matter search results from 4.2 tonne-years of exposure of the LUX-ZEPLIN (LZ) experiment*, Physical Review Letters **135**, 011802 (2025); arXiv:2410.17036.

[15] E. Aprile *et al.* (XENON Collaboration), *First dark matter search with nuclear recoils from the XENONnT experiment*, Physical Review Letters **131**, 041003 (2023); arXiv:2303.14729.

[16] Z. Bo *et al.* (PandaX Collaboration), *Dark matter search results from 1.54 tonne·year exposure of PandaX-4T*, Physical Review Letters **134**, 011805 (2025); arXiv:2408.00664.

[17] P. Agnes *et al.* (DarkSide-50 Collaboration), *Search for low-mass dark matter WIMPs with 12 ton-day exposure of DarkSide-50*, Physical Review D **107**, 063001 (2023); arXiv:2207.11966.